

ANKITA DAS UG/02/BTCSECSF/2023/017

UDB LAB 7

Data:

```
{ roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },  
{ roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },  
{ roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },  
{ roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },  
{ roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },  
{ roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },  
{ roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },  
{ roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }.
```

```
db.student_UG_02_BTCSEAIML_2023_063.insertMany([  
{ roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },  
{ roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },  
{ roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },  
{ roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },  
{ roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },  
{ roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },  
{ roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },  
{ roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }  
])
```

```

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ca16b903512a77c544152e'),
    '1': ObjectId('69ca16b903512a77c544152f'),
    '2': ObjectId('69ca16b903512a77c5441530'),
    '3': ObjectId('69ca16b903512a77c5441531'),
    '4': ObjectId('69ca16b903512a77c5441532'),
    '5': ObjectId('69ca16b903512a77c5441533'),
    '6': ObjectId('69ca16b903512a77c5441534'),
    '7': ObjectId('69ca16b903512a77c5441535')
  }
}
[
  {
    _id: ObjectId('69c0e639752b6f69fd44152e'),
    roll: 1,
    name: 'Anu',
    age: 22,
    department: 'IT',
    marks: 90,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e639752b6f69fd44152f'),
    roll: 2,
    name: 'Riya',
    age: 21,
    department: 'CSE',

```

Q1. Display all documents. db.student_UG_02_BTCSECSF017.find()

```

},
{
  _id: ObjectId('69c0e639752b6f69fd441530'),
  roll: 3,
  name: 'Aman',
  age: 23,
  department: 'IT',
  marks: 92,
  status: 'Pass'
},
{
  _id: ObjectId('69c0e639752b6f69fd441531'),
  roll: 4,
  name: 'Neha',
  age: 20,
  department: 'ECE',
  marks: 74,
  status: 'Pass'
},
{
  _id: ObjectId('69c0e639752b6f69fd441532'),
  roll: 5,
  name: 'Raj',
  age: 24,
  department: 'CSE',
  marks: 88,
  status: 'Pass'
},

```

Q2. Find students from IT department db.student_UG_02_BTCSE_CSF017.find({
department: "IT" })

```

type: 'UG', roll: 1, marks: 90, status: 'Pass'
},
{
  _id: ObjectId('69c0e639752b6f69fd441530'),
  roll: 3,
  name: 'Aman',
  age: 23,
  department: 'IT',
  marks: 92,
  status: 'Pass'
},
{
  _id: ObjectId('69c0e639752b6f69fd441532'),
  roll: 5,
  name: 'Raj',
  age: 24,
  department: 'CSE',
  marks: 88,
  status: 'Pass'
},
{

```

Q3. Find students with marks > 80

db.student_UG_02_BTCSE_CSF017.find({ marks: { \$gt: 80 } })

```

{
  _id: ObjectId('69c0e639752b6f69fd441533'),
  roll: 6,
  name: 'Simran',
  age: 22,
  department: 'IT',
  marks: 81,
  status: 'Pass'
},
{
  _id: ObjectId('69c0e639752b6f69fd441535'),
  roll: 8,
  name: 'Priya',
  age: 23,
  department: 'CSE',
  marks: 95,
  status: 'Pass'
},
{
  _id: ObjectId('69c0e683752b6f69fd441536'),
  roll: 1,
  name: 'Anu',
  age: 22,
  department: 'IT',
  marks: 85,
  status: 'Pass'
},
{

```

Q4. Insert one new student db.student_UG_02_BTCSE_CSF017_063.insertOne(

```

{ roll: 9, name: "Rahul", age: 22, department: "IT", marks: 83 }
)

```

```
{
  _id: ObjectId('69c0e685752b6f69fd44153e'),
  roll: 9,
  name: 'Rahul',
  age: 22,
  department: 'IT',
  marks: 83,
  status: 'Pass'
},
```

Q5. Update marks of Anu to 90.

```
db.student_UG_02_BTCSE_CSF017.updateOne(
{ name: "Anu" },
{ $set: { marks: 90 } }
)
```

```
{
  _id: ObjectId('69c0e639752b6f69fd44152e'),
  roll: 1,
  name: 'Anu',
  age: 22,
  department: 'IT',
  marks: 90,
  status: 'Pass'
},
```

Q6. Add a new field "status: Pass" for all students db.student_UG_02_BTCSE

```
_CSF017.updateMany(
{ },
{ $set: { status: "Pass" } }
)
```

```
{
  _id: ObjectId('69c0e639752b6f69fd44152f'),
  roll: 2,
  name: 'Riya',
  age: 21,
  department: 'CSE',
  marks: 78,
  status: 'Pass'
},
{
  _id: ObjectId('69c0e639752b6f69fd441530'),
  roll: 3,
  name: 'Aman',
  age: 23,
  department: 'IT',
  marks: 92,
  status: 'Pass'
},
{
  _id: ObjectId('69c0e639752b6f69fd441531'),
  roll: 4,
  name: 'Neha',
  age: 20,
  department: 'ECE',
  marks: 74,
  status: 'Pass'
},
}
```

Q7. Delete all students with marks < 70

```
db.student_UG_02_BTCSE_CSF017.deleteMany(
{ marks: { $lt: 70 } }
)
```

```
Type "it" for more
{
  acknowledged: true,
  insertedId: ObjectId('69ca16b903512a77c5441536')
}
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 62,
  modifiedCount: 9,
  upsertedCount: 0
}
{ acknowledged: true, deletedCount: 1 }
24
```

Q8. Count students in CSE department db.student_UG_02_BTCSE_CSF017.countDocuments(

```
{ department: "CSE" }
)
```

Q9. Find total students in each department `db.student_UG_02_BTCSE_CSF017.aggregate([`

`{ $group: { _id: "$department", total: { $sum: 1 } } }`

`])`

```
[
  { _id: 'CSE', total: 33 },
  { _id: 'IT', total: 41 },
  { _id: 'ECE', total: 11 }
]
```

Q10. Find highest marks `db.student_UG_02_BTCSE_CSF017.find()`

`.sort({ marks: -1 })`

`.limit(1)`

```
[
  {
    _id: ObjectId('69c0e639752b6f69fd441535'),
    roll: 8,
    name: 'Priya',
    age: 23,
    department: 'CSE',
    marks: 95,
    status: 'Pass',
    grade: 'A'
  }
]
```

Q11. Find the second highest marks. `db.student_UG_02_CSE_CSF017.find()`

`.sort({ marks: -1 })`

`.skip(1)`

`.limit(1)`

```
[
  {
    _id: ObjectId('69c0e639752b6f69fd441535'),
    roll: 8,
    name: 'Priya',
    age: 23,
    department: 'CSE',
    marks: 95,
    status: 'Pass',
    grade: 'A'
  }
]
```

Q12. Find students whose name starts with 'A' and marks > 80 `db.student_UG_02_BTCSE_CSF017.find({`

```
name: { $regex: "^A" }, marks:
{ $gt: 80 }

})
```

```
    grade: 'A'
  },
  {
    _id: ObjectId('69ca188d03512a77c5441540'),
    roll: 1,
    name: 'Anu',
    age: 22,
    department: 'IT',
    marks: 85,
    status: 'Pass',
    grade: 'B'
  },
  {
    _id: ObjectId('69ca188d03512a77c5441542'),
    roll: 3,
    name: 'Aman',
    age: 23,
    department: 'IT',
    marks: 92,
    status: 'Pass',
    grade: 'A'
  }
]
```

Q13. Display department-wise average marks, only for departments where avg > 80

```
db.student_UG_02_BTCSE_CSF017.aggregate([

{ $group: { _id: "$department", avgMarks: { $avg: "$marks" } } },

{ $match: { avgMarks: { $gt: 80 } } }

])
```

```
[
  { _id: 'IT', avgMarks: 85.53658536585365 },
  { _id: 'CSE', avgMarks: 87 }
]
```

Q14. Find students with marks between 75 and 90, sorted by marks ascending

```
db.student_UG_02_BTCSE_CSF017.find(

{ marks: { $gte: 75, $lte: 90 } }
```

`).sort({ marks: 1 })`

```
[
  {
    _id: ObjectId('69c0e639752b6f69fd44152f'),
    roll: 2,
    name: 'Riya',
    age: 21,
    department: 'CSE',
    marks: 78,
    status: 'Pass',
    grade: 'C'
  },
  {
    _id: ObjectId('69c0e683752b6f69fd441537'),
    roll: 2,
    name: 'Riya',
    age: 21,
    department: 'CSE',
    marks: 78,
    status: 'Pass',
    grade: 'C'
  },
]
```

```
{
  _id: ObjectId('69ca16b903512a77c544152f'),
  roll: 2,
  name: 'Riya',
  age: 21,
  department: 'CSE',
  marks: 78,
  status: 'Pass',
  grade: 'C'
},
{
  _id: ObjectId('69c0e6a5752b6f69fd441544'),
  roll: 6,
  name: 'Simran',
  age: 22,
  department: 'IT',
  marks: 81,
  status: 'Pass',
  grade: 'B'
},
]
```

Q15. Add grade field based on marks

`db.student_UG_02_BTCSE_CSF017.updateMany(`

```
{},
[
  { $set: {
    grade: {
      $switch: {
        branches: [
```



```
{ case: { $gte: ["$marks", 90] }, then: "A" }, {  
  case: { $gte: ["$marks", 80] }, then: "B" },  
  
  { case: { $gte: ["$marks", 70] }, then: "C" }  
],  
default: "D"  
}  
}  
}  
}]  
)
```

```
Type "it" for more  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 85,  
  modifiedCount: 0,  
  upsertedCount: 0  
}
```